

Investigating Postoperative Hypoxemia Using the Medical Informatics Operating Room Vitals and Events Repository (MOVER)

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Abstract

Introduction & Literature Review

- Postoperative hypoxemia is a frequent complication from anesthetic¹
- The MOVER EPIC dataset includes records from 64,354 surgeries between 2017 and 2022² and can be used to study this issue
- Prior research has assessed how postoperative hypoxemia is affected by different variables, but these models only contained up to four laboratory test results^{3,4,5}
- To fill this gap characterized by few laboratory test results, this work modelled postoperative hypoxemia probability using logistic regression with up to ten preoperative laboratory test results and three demographic variables as covariates
- The objective was to detect significant predictors of postoperative hypoxemia and determine the accuracy of predicting its probability with just these variables

Methodology (The Pipeline)

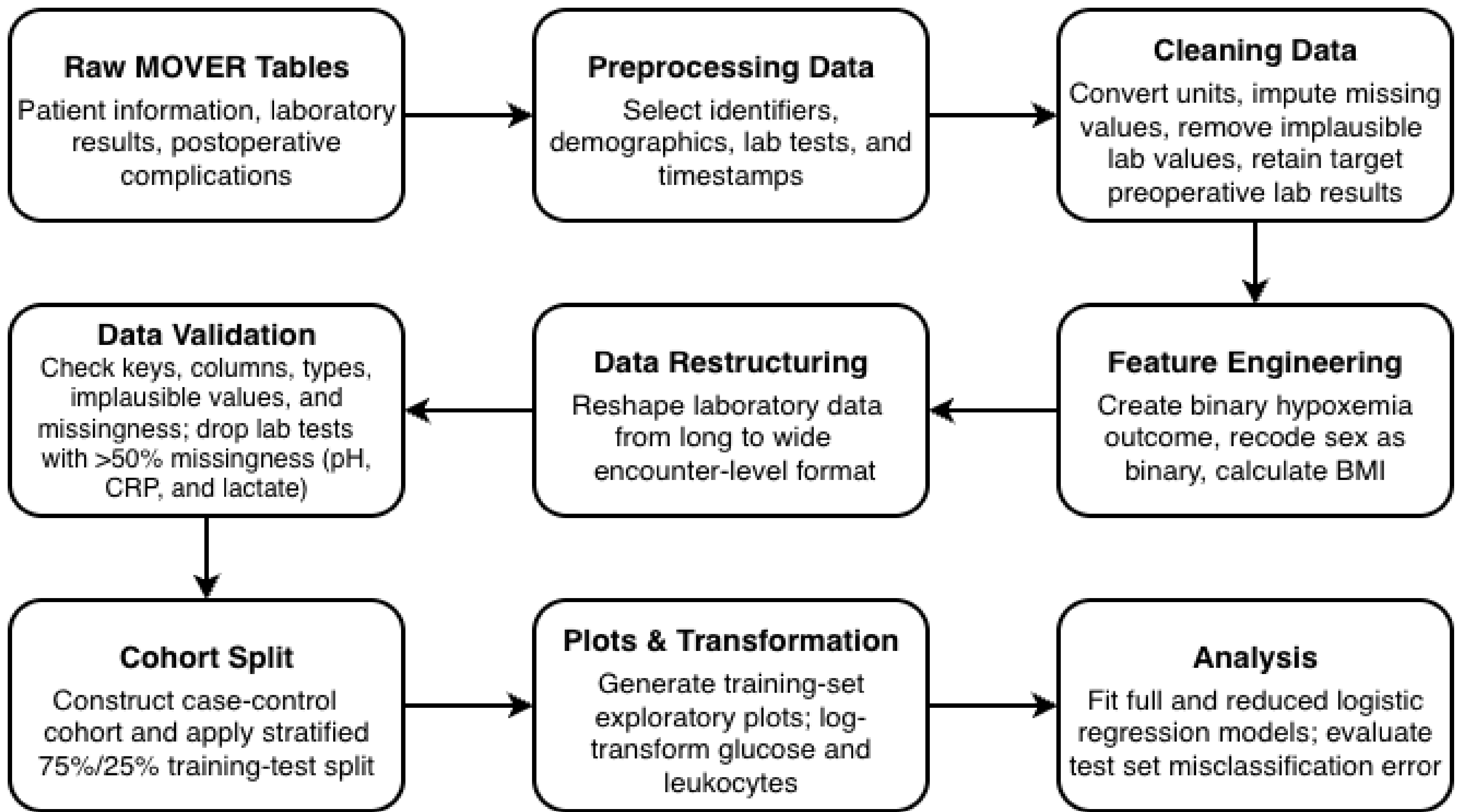


Figure 1: Flow chart of data wrangling architecture

Key Findings

Conclusion & Impact

Key Takeaway: Male patients with high BMI and high glucose may benefit from closer postoperative monitoring for early detection and timely intervention of hypoxemia.

Key Risk Factors

- **Higher BMI** elevates odds of postoperative hypoxemia , aligning with established literature⁴
- **Higher glucose level** were significantly associated with increased odds of hypoxemia
- **Male** patients have higher odds of hypoxemia, which contradicts previous findings that showed being female as a risk factor⁴

Model Performance and Limitation

- The full model achieved a lower test misclassification rate (37%) than the reduced model (43%), suggesting that including a broader set of lab test results is necessary to capture the complexity of hypoxemia
- Only [percentage]% of patients reported postoperative hypoxemia, likely contributing to the relatively high misclassification rate

Future Directions

- Employing oversampling or re-weighting techniques to better account for imbalance and improve predictive accuracy
- Incorporating a wider variety of laboratory tests to strengthen predictive power
- Exploring how hypoxemia interacts with other concurrent postoperative complications

References

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